

2018 RI H2 Chemistry Preliminary Examinations Paper 4 – Preparation List

Apparatus list (per student)

s/n	Apparatus	Quantity	Question Number
1	50 cm ³ burette	2	1
2	Filter funnel	2 (1 glass, 1 plastic)	1 and 3
3	25 cm ³ Pipette	1	1
4	Pipette filler	1	1
5	White tile	1	1
6	250 cm ³ conical flask	2	1
7	250 cm ³ volumetric flask	1	1
8	10 cm ³ measuring cylinder	1	1 and 3
9	Deionised water bottle	1	1, 2 and 3
10	100 cm ³ beaker	1	1
11	250 cm ³ beaker	1	2
12	50 cm ³ measuring cylinder	2	2
13	Polystyrene cup	2	2
14	Thermometer	1	2
15	Test-tube rack	1	3
16	Test-tubes	4	3
	Boiling tube	2	2
	Droppers	4	1,2 and 3
17	Bunsen burner, shield and rubber mat	1	3
18	Lighter	1	3
19	Glass rod	1	3
20	Wooden splint, litmus papers and filter paper	—	3

Chemical List (per student)

Code	Identity of reagent	Question	Quantity required
FA 1	0.05 mol dm ⁻³ Na ₂ S ₂ O ₃	1	120 cm ³
FA 2	0.276 mol dm ⁻³ CuSO ₄	1,3	80 cm ³
FA 3	10% KI	1	40 cm ³
Starch	2% starch solution	1	5 cm ³
FA 5	0.25 mol dm⁻³ sodium chlorate(I), NaClO 10% NaClO, Density = 1.10g/ml MW=74.44g/mol Molarity = (10%/100)x1.10x(1/74.4)(1000) = 1.478 mol/L m1v1=m2v2 v1 = (1000x0.25)/1.478 = 169.15 ml/L	2	160 cm ³
FA 6	0.100 mol dm ⁻³ Na ₂ S ₂ O ₃ in 0.08 mol dm⁻³ NaOH Weigh the salts and mixed. Make-up to 1 dm ³ .	2	160 cm ³
FA 8	Zn powder	3	B155 and spoon
FA 9	0.5 mol dm ⁻³ ammonium vanadate(V), NH ₄ VO ₃ (aq) Weigh the chemical and dissolves into 2M NaOH per each dm ³ .	3	10 cm ³

Bench reagents

1. $1 \text{ mol dm}^{-3} \text{ H}_2\text{SO}_4$
2. 2M NH_3
3. $\text{Ca(OH)}_2(\text{aq})$
4. NH_4Cl solid (8g each)
5. Aq. H_2O_2 3% solute